DAY 16 — Model Evaluation Metrics & Error Analysis

Day 16 covered one of the most critical aspects of any machine learning workflow: **Model Evaluation**. Training a model is only half the job — evaluating it correctly determines whether the model is actually performing well or just memorizing patterns. Today's session focused on understanding error metrics, comparing predictions, and analyzing model reliability.

🔹 1. Importance of Evaluation Metrics

Machine learning models must be evaluated because:

- They may overfit or underfit
- Predictions may look accurate but have high error
- Business decisions depend on trustworthy metrics
- Comparative analysis requires numerical benchmarks

Good evaluation makes the model more reliable and suitable for real-world deployment.

🔸 2. Key Regression Metrics

Since stock prediction is a **regression problem**, we studied:

✓ Mean Absolute Error (MAE)

Average difference between predictions and actual values.
Simple and easy to interpret.

✓ Mean Squared Error (MSE)

Penalizes larger errors more heavily (good for financial forecasting).

✓ Root Mean Squared Error (RMSE)

Square root of MSE; provides error in the same unit as the target.

✓ R-Squared (R^2 Score)

Explains how much variance the model captures.

High R^2 = More accurate model.

🔸 3. Error Distribution Analysis

We plotted:

- Error histogram

- Actual vs predicted scatter plot
- Error line plot

These helped identify:

- Bias patterns
- Sudden spikes
- Inconsistent predictions

Prophet handled uncertainty well, making errors stable across the timeline.

🔸 4. Cross-Validation Techniques

We learned:

- Train-test split
- Rolling window validation
- K-fold cross-validation

Time-series datasets require **time-aware validation** to avoid data leakage.

🔸 5. Evaluating Prophet Forecasts

Prophet's internal metrics include:

- Trend accuracy
- Seasonality accuracy
- Prediction confidence intervals

Stock forecasting is unpredictable due to:

- News events
- Market crashes
- Volatility

Therefore, Prophet's uncertainty intervals were extremely valuable.

Key Takeaways from Day 16

- Metrics give numerical clarity on model performance
- RMSE is preferred for financial models
- Error visualizations show deep insights

- Cross-validation ensures model stability
- Evaluation determines if the model is ready to deploy